Python Fundamentals MCQ Assessment

50 Questions • Variables, Control Flow, Collections, Loops & Strings

Total Questions: 50
Time Allowed: 60 Mins

PART 1: VARIABLES, DATA TYPES & OPERATORS

Q1. Which of the following is an invalid variable name in Python?

A) _my_var
B) 2my_var
C) my_var2
D) myVar

Q2. What is the output of `type(10 / 2)` in Python 3?

A) <class 'int'>
B) <class 'float'>
C) <class 'double'>
D) <class 'number'>

Q3. What is the result of the expression `2 ** 3 ** 2`?

A) 64
B) 512
C) 128
D) 256

Q4. Which operator performs floor division?

A) /
B) //
C) %
D) div

Q5. What will `bool("False")` evaluate to?

A) False
B) True
C) ValueError
D) None

Q6. Which data type is immutable in Python?

A) List
B) Dictionary
C) Tuple
D) Set

Q7. What is the output of `10 % 3`?

A) 3
B) 1
C) 3.33
D) 0

Q8. What is the result of `True and not False`?

A) False
B) True
C) None
D) Error

Q9. What does the `is` operator evaluate?

A) Value equality
B) Identity reference
C) Data type
D) Length

Q10. What is the output of `x = 5; x += 3 * 2`?

A) 16
B) 11
C) 13
D) 25

PART 2: STRING OPERATIONS & SLICING

Q11. What is the output of `"Python"[::-1]`?

A) Python
B) nohtyP
C) P
D) Error

Q12. Which string method removes trailing whitespace?

A) strip()
B) lstrip()
C) rstrip()
D) trim()

PART 2: STRING OPERATIONS (CONTD.) & CONTROL FLOW

Q13. What is the output of `"a" + "b" * 2`?

A) abab
B) abb
C) a2b2
D) TypeError

Q14. What does `"hello".find("z")` return?

A) 0
B) -1
C) ValueError
D) None

Q15. How do you format variables inside strings in Python 3.6+?

A) f"Hello {name}"
B) "Hello %s" % name
C) "Hello {}".format(name)
D) All of the above

Q16. What is the output of `len("Hello\nWorld")`?

A) 10
B) 11
C) 12
D) 5

PART 3: CONTROL FLOW & CONDITIONAL STATEMENTS

Q17. Which keyword is used for multi-way conditional branching?

A) else if
B) elseif
C) elif
D) switch

Q18. What is printed?

```
x = 10 if x > 5: print("A") elif x > 8: print("B") else: print("C")
```

A) A
B) B
C) A and B
D) C

Q19. Which value evaluates to `False` in a condition?

A) [] (empty list)
B) 0
C) None
D) All of the above

Q20. What is the output of `x = 5 if 10 > 20 else 10`?

A) 5
B) 10
C) True
D) False

Q21. Which Python 3.10+ feature replaces traditional `switch-case`?

A) select case
B) match case
C) do condition
D) branch

Q22. What happens if no condition is True in an `if-elif` chain with no `else`?

A) SyntaxError
B) Nothing executes
C) Raises exception
D) Returns None

PART 4: LOOPS & ITERATION CONTROL

Q23. **What is generated by `list(range(1, 5))`?**

A) [1, 2, 3, 4, 5]
B) [1, 2, 3, 4]
C) [0, 1, 2, 3, 4]
D) [2, 3, 4, 5]

Q24. **What does `break` statement do inside a loop?**

A) Skips current step
B) Terminates loop
C) Restarts loop
D) Exits program

Q25. **When does a loop's `else` block execute?**

A) When break occurs
B) When loop finishes normally
C) Always
D) Never

Q26. **What is the output?**

```
for i in range(3): if i == 1: continue
print(i, end=" ")
```

A) 0 1 2
B) 0 2
C) 1 2
D) 0 1

Q27. **How to iterate with index and value simultaneously?**

A) zip()
B) enumerate()
C) range()
D) items()

Q28. **What is the output of `list(range(5, 0, -2))`?**

A) [5, 3, 1]
B) [5, 3, 1, -1]
C) [5, 4, 3, 2, 1]
D) []

PART 5: LISTS & DYNAMIC ARRAYS

Q29. **Which method adds an element to the end of a list?**

A) add()
B) insert()
C) append()
D) extend()

Q30. **What is the output?**

```
a = [1, 2] b = a b.append(3) print(a)
```

A) [1, 2]
B) [1, 2, 3]
C) [3]
D) Error

Q31. **Difference between `pop()` and `remove()`?**

A) pop removes by index; remove by value
B) pop removes by value; remove by index
C) Both do identical operations
D) pop is for tuples only

Q32. **What is the output of `[x*2 for x in range(3)]`?**

A) [0, 2, 4]
B) [2, 4, 6]
C) [0, 1, 2]
D) [1, 2, 3]

PART 6: TUPLES & DICTIONARIES

Q33. How do you declare a single-element tuple?

A) t = (5)
B) t = (5,)
C) t = tuple(5)
D) t = [5]

Q34. What happens on `t = (1, 2); t[0] = 5`?

A) Tuple becomes (5, 2)
B) TypeError
C) ValueError
D) IndexOutOfBounds

Q35. Can a dictionary key be a list?

A) Yes, always
B) No, lists are unhashable
C) Only if empty
D) Yes, with tuple conversion

Q36. What is the output?

```
d = {"a": 1, "b": 2} print(d.get("c", 3))
```

A) KeyError
B) None
C) 3
D) 0

Q37. Which method returns dictionary key-value pairs?

A) keys()
B) values()
C) items()
D) pairs()

Q38. Output of `len({1, 1, 2, 3, 3})`?

A) 5
B) 3
C) 2
D) Error

Q39. What is the result of `d = {}; type(d)`?

A) <class 'dict'>
B) <class 'set'>
C) <class 'tuple'>
D) <class 'list'>

Q40. How to merge dictionary `b` into dictionary `a` in Python 3.9+?

A) a + b
B) a | b
C) a & b
D) merge(a, b)

PART 7: COMBINED APPLIED CONCEPTS

Q41. What is the output?

```
x = [1, 2, 3] y = x.copy() print(x is y)
```

A) True
B) False
C) Error
D) None

Q42. What is the default return value of a function without a return statement?

A) 0
B) False
C) None
D) Null

PART 7: COMBINED APPLIED CONCEPTS (CONTD.)

Q43. What is the output of `"a.b.c".split(".")`?

A) ['a', 'b', 'c']
B) ("a", "b", "c")
C) "abc"
D) ['a.b.c']

Q44. Result of `list(set([1, 2, 2, 3, 1]))`?

A) [1, 2, 2, 3, 1]
B) [1, 2, 3]
C) (1, 2, 3)
D) {1, 2, 3}

Q45. Output of `bool([])`?

A) True
B) False
C) None
D) Error

Q46. What does `pass` statement do?

A) Skips loop execution
B) Null placeholder operation
C) Exits current block
D) Returns True

Q47. What is the output?

```
a = (1, [2, 3]) a[1].append(4) print(a)
```

A) TypeError
B) (1, [2, 3, 4])
C) (1, [2, 3])
D) AttributeError

Q48. Result of `":".join(["A", "B", "C"])`?

A) "A:B:C"
B) "ABC:"
C) [":A", ":B", ":C"]
D) TypeError

Q49. What is the output?

```
count = 1 while count < 4: count += 1
print(count)
```

A) 3
B) 4
C) 5
D) 1

Q50. Which operator is used for bitwise AND?

A) &&
B) and
C) &
D) |

OFFICIAL ANSWER KEY

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	**B**	11	**B**	21	**B**	31	**A**	41	**B**
2	**B**	12	**C**	22	**B**	32	**A**	42	**C**
3	**B**	13	**B**	23	**B**	33	**B**	43	**A**
4	**B**	14	**B**	24	**B**	34	**B**	44	**B**
5	**B**	15	**D**	25	**B**	35	**B**	45	**B**
6	**C**	16	**B**	26	**B**	36	**C**	46	**B**
7	**B**	17	**C**	27	**B**	37	**C**	47	**B**
8	**B**	18	**A**	28	**A**	38	**B**	48	**A**
9	**B**	19	**D**	29	**C**	39	**A**	49	**B**
10	**B**	20	**B**	30	**B**	40	**B**	50	**C**